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THE FACTOR OF GENERAL ORIENTATION IN MAZE LEARNING IN THE WHITE RAT*

From the Animal Laboratory of the Department of Psychology, Columbia University

C. J. WARDEN AND F. R. FOX[1]

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In much of the early literature on maze learning in the white rat it would seem that the major emphasis was placed upon the fixation of the several elements of the pattern, while the more general factors involved were passed over rather lightly. Attention was first directed specifically to the factor of general orientation by Peterson (5, 6) in connection with his "completeness of response" principle. Further work bearing upon the importance of general orientation in maze learning has been reported by Hubbert and Lashley (4), Dashiell (2, 3), and Warden (7). In the main, the evidence presented in these later studies is drawn from an analysis of error data in which it is found that the relative difficulty of various units of the pattern is related more or less directly to the appearance of the tendency to orient toward the food box. This is true not merely of backward errors, but, as each of these studies shows, of certain types of forward culs-de-sac errors as well. The present report, covering an investigation conducted in 1927-1928, offers further evidence on this point in connection with the several experimental conditions to be described.

TEST 1

The first test deals with the influence of various types of rotation after the maze habit has been established upon the locus of the errors made during later maze performance. The apparatus used was a 5 culs-de-sac pattern, as shown in the diagram of Figure 1, arranged from units of the Warner-Warden maze (8). White rats, approximately 75 days of age, and weighing 72 grams, on the average, were employed. The apparatus was set up in the room in which the rat

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[1]This experiment was planned and carried out in cooperation, the junior author securing and tabulating the data, and the senior author being responsible for the report in its present form.

colony was housed. The fact that the cages were all on one side of the maze suggested the possibility that noise from the colony might serve as a secondary cue. This does not seem likely, however, since no disturbance resulted when the colony cages were moved 90° to the right and placed the same distance from the maze. A radiator stood several feet away on one side of the maze, and a check on possible cues from this source was had by turning the radiator off and placing electric stoves on the opposite side of the apparatus. Since this had no apparent influence on the performance of the animals, it is likely that the radiator, which was kept at a much lower temperature than the stoves, was also ineffective.

Each animal was given one preliminary trial per day for 10 days,

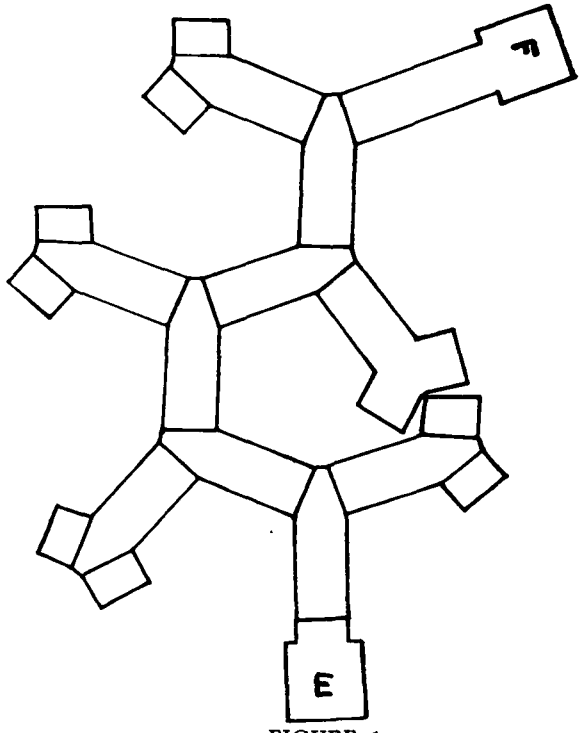


FIGURE 1

FIVE CULS-DE-SAC PATTERN OF WARNER-WARDEN MAZE

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during which it was allowed to explore the outside of the maze and, after locating the food box, to feed for 10 seconds. If the animal failed to locate the food box within 2 minutes it was carried there by the experimenter. This procedure doubtless considerably reduced the random exploration of the early trials, since in no case during the regular training series did an animal take longer than 90 seconds to reach the food box. It probably had the effect also of preventing backward errors in the later training. After feeding for 10 seconds, the animal was placed in the feeding cage with access to whole-wheat bread soaked in milk for 30 minutes.

One trial per day was given in the training series proper, the animal being allowed a nibble of moistened McCollum's diet after reaching the food box, and then being transferred to the feeding cage as in the preliminary trials. An error was checked whenever the rat extended its nose into a cul-de-sac, only culs-de-sac errors being scored. Except in the case of Groups C and D, in which overlearning was specifically intended, the norm of mastery required in the initial learning was always 4 successive errorless trials (Norm A). In these two groups the norm set was 27 errorless trials out of 30 (Norm B). In the relearning series following rotation, Norm A was employed throughout, except in the case of Groups E and F in which no norm whatever was used, data being taken only on the

TABLE 1
A SUMMARY OF GROUPS AND CONDITIONS

Group	Size of Group	Initial Learning Conditions	Relearning Conditions
A	9	Norm A	Maze rotated 90° to the right; relearning to norm A.
B	4	Norm A	Cage and maze rotated 90° to the right about common axis; relearning to Norm A.
C	4	Norm B	Same as in B except animals caged in new position 48 hours before relearning began.
D	4	Norm B	Maze rotated 90° to right after overlearning; relearning to Norm A.
E	3	Norm A	Dizziness induced by 70 revolutions in 20 sec. in drum; maze not rotated; disturbance on 4 trials noted.
F	3	Norm A	Same as in E, except maze rotated 90° to right in addition to dizziness; disturbance on 4 trials noted.

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TABLE 2

SHOWING INFLUENCE OF ROTATION AND OTHER FACTORS ON MAZE PERFORMANCE, INTRODUCED AFTER LEARNING HAS BEEN COMPLETED

Group	Trials Av.	Trials *M.V.*	Time in Secs. Av.	Time in Secs. *M.V.*	Time in Secs. Final	Errors Av.	Errors *M.V.*	Percentage error scores for culs-de-sac 1	2	3	4	5
A	5.2	1.4	67.4	19.6	3.2	10.6	3.7	4.4	2.3	1.8	1.0	0.5
	5.2	3.2	52.0	20.8	3.2	9.6	3.3	5.2	1.2	0.9	2.1	0.6
B	3.0	0.0	86.3	22.0	4.0	8.5	3.0	4.1	1.9	2.0	2.0	0.0
	4.5	1.0	47.8	17.6	3.6	10.0	2.5	5.3	1.4	1.0	2.0	0.3
C	5.5	2.3	68.8	7.2	2.6	9.5	2.0	4.9	1.5	2.6	0.7	0.3
	3.0	1.5	36.5	20.5	3.0	5.3	2.8	5.9	1.7	1.6	0.5	0.3
D	3.5	0.8	60.5	5.9	2.7	8.5	0.8	4.0	2.4	1.6	1.2	0.8
	3.5	0.8	66.6	13.6	2.8	10.0	2.0	4.1	2.5	2.2	0.9	0.3
E	9.0	2.7	105.1	26.3	4.8	12.2	2.0	6.6	1.9	1.1	0.0	0.4
	4 only		69.3	18.2	. .	6.3	0.8	. .	. .	. .	. .	. .
F	4.3	0.4	74.7	34.2	4.0	7.0	2.7	7.0	1.2	0.7	1.1	0.0
	4 only		67.4	16.8	. .	6.3	2.0	. .	. .	. .	. .	. .

first 4 trials following rotation. Time per trial was taken with a stop-watch throughout.

The several conditions investigated are indicated in Table 1. The initial learning and the relearning after rotation was continuous for each animal, except that an extra day was allowed to intervene in the case of Group C in order to allow a 48-hour period of adaptation to the rotated cage and maze before beginning to relearn the latter in its new position. As will be seen, all of the groups except the first are relatively small, and this fact must be taken into account in drawing conclusions regarding the influence of the various conditions introduced.

The results secured in this experiment are presented in Table 2. In addition to the total time score for each group, as shown in the third column, the "final" score covering the time record of the four perfect runs of the norm of mastery is given in the fifth column. The scores showing the distribution of errors in the several culs-de-sac are given in percentages in order to make them more strictly comparable from group to group. The record of Group A, in which the usual method of maze rotation was applied, may be taken as basic to a comparison of the results obtained when other conditions were introduced.

It is interesting to note that the influence of rotation was not

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significantly reduced when both cage and maze were rotated together about their common axis, as in the case of Group B. In both the initial learning and the relearning tests, the animals were carried from the cage to the entrance box in a straight line and placed in the food box with forward orientation. The animals had been transferred to the new cage position 24 hours before the first trial was given, or immediately after the last trial of the initial training series. It was thought that an extra day in the new position might reduce any possible emotional disturbance, but this change in procedure, in the case of Group C, did not affect the rotation scores markedly, and no conclusion seems to be warranted on account of the small size of the group. The results for Groups C and D, in which Norm B was used in the initial learning, seem to indicate that rotation effects are not influenced to any great extent by the overlearning represented.

In the case of Group E, dizziness was substituted for rotation. The dizziness was induced by rotating the animals singly in a revolving-drum activity cage turned on the side so that the axis was vertical. Approximately 70 revolutions of the wheel in 20 seconds was sufficient to make the animals so dizzy that, when placed in the entrance box, they became rigid with limbs outstretched. After a time they would suddenly jump up with a frog-like jerk and approach the door of the maze, often moving in a small circle. The direction of revolving the wheel was determined by chance. The amount of disturbance on running the maze under these conditions appears to be about the same as that which occurred in general from rotation effects, insofar as can be judged from the record of the first four trials. When both dizziness and maze rotation was introduced, as was true in the case of Group F, the resulting disturbance did not seem to be greater than when the animals were made dizzy and the maze not rotated, as with Group E.

The findings enumerated above are in general agreement with the results obtained by others in rotation work on the white rat. Our primary interest, however, was not so much in the effect of rotation itself as in the possible transfer of tendencies established during the initial training to the rotation situation. It seems highly probable that a general orientation tendency set up during the initial training would itself become a disturbing factor after rotation, since the absolute direction of the food box would be changed. The results seem to support this presumption. In Group A, for example, the percent-

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age of errors in the several culs-de-sac shows a fairly regular decrease from the first one to the fifth, in the learning series. In the relearning series, however, the order is 1, 4, 2, 3, 5, which means that culs-de-sac 2 and 3 were entered less often and culs-de-sac 1 and 4 significantly more often than during the initial learning. The most natural explanation would seem to be that the latter two culs-de-sac, lying on the same side of the pathway, are located after rotation in the general direction of approach to the old position of the food box. This shift in the locus of errors is in agreement with expectation on the supposition that the general orientation tendency of the initial learning is carried over, in some measure, to the new situation. The shift is less clearly indicated in the results of Groups B, C, and D, which were smaller in size. That the tendency appears to be present to some extent throughout, even though other conditions were being varied from group to group, suggests that it may be a factor of considerable significance in normal learning situations as well.

Test 2

In this test, two patterns of a modified Y-maze, the turns in the one pattern being the reverse of those in the other, were employed. These were arranged from units of the Warner-Warden maze (8), the ground plan of both patterns being shown in Figure 2. A total of 16 male white rats, weighing approximately 110 grams, were tested. They were placed singly in the entrance box for 30 seconds, then transferred to the food box and allowed to feed for 10 seconds on each of three days before beginning the training. One trial a day was given during the training proper until the norm of three successive perfect trials was reached. On the day following, they were given one trial only in the maze with turns reversed. The scores covering the learning period were as follows: average trials, 3.1 ± 0.9; average total time, 10.3 ± 4.7; average time of last trial, 3.0 ± 0.42.

Exactly 50% of the animals entered the cul-de-sac on the first trial of the initial learning series. In the one trial given on the reversed pattern, only two animals entered the cul-de-sac, which shows that the general orienting tendency did not carry over under the present condition in any effective manner. It should be noted, however, that the time score showed a marked increase under the new condition, the average time being 4.7 seconds as against 3.0

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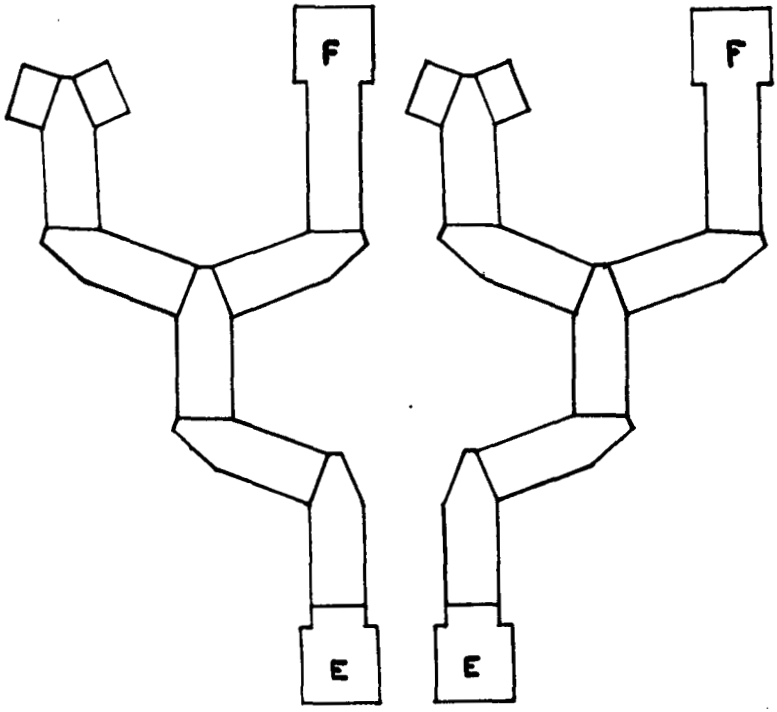


FIGURE 2
MODIFIED Y-MAZE
The original pattern is shown on the left and the reversed pattern on the right.

seconds on the last trial of the training series. The time was very much longer in the case of the two rats which entered the cul-de-sac than for the other 14 animals, the score being 7.4 seconds for one and 12.0 seconds for the other, while the average score for the 14 successive animals was only 4.0 seconds.

TEST 3

The aim of this test was to determine whether or not the white rat can learn, under carefully controlled conditions, to turn consistently to the north in reaching the food box. The apparatus employed was a simple T-maze, with wooden sides 5.5 inches high and a mesh top, the pathway being 5 inches wide. The pathway leading to the

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TABLE 3
SHOWING RESPONSES TO GOAL REQUIRING TURN TO NORTH.
R and *L*, right and left; *N* and *S*, north and south.

Trial	North position	Rat 1	Rat 2	Rat 3	Rat 4
1	R	N R	N R	S L	N R
2	L	S R	S R	S R	N L
3	R	S L	S L	N R	S L
4	L	N L	S R	N L	N L
5	L	N L	S R	N L	N L
6	R	N R	S L	S L	S L
7	L	S R	N L	N L	N L
8	R	S L	N R	S L	S L
9	R	N R	N R	S L	N R
10	L	N L	S R	N L	N L
11	L	S R	N L	N L	N L
12	R	N R	N R	S L	S L
13	L	S R	S R	N L	N L
14	R	N R	N R	S L	S L
15	R	N R	N R	S L	S L
16	L	S R	S R	N L	N L
17	R	N R	N R	S L	S L
18	L	S R	S R	N L	N L
19	L	S R	S R	N L	N L
20	R	N R	N R	S L	S L
21	rot. 90°	R	R	L	L
22	rot. 90°	R	R	L	L

arms, and the arms themselves, were 15 inches long. At a point 11 inches inward from the end of each arm a partition, 2.5 inches high, was placed, which kept the food out of sight of the animal until this partition had been crawled over. Food was placed at the end of both so as to equalize the odor conditions, but the food in the south arm, or cul-de-sac, was always enclosed in a wire-mesh box so that it was not available to the animal. In order to eliminate visual cues from the environment, the table on which the maze rested was screened over with double-ply black satin so as to form a dark chamber 36 x 36 x 40 inches, this being lighted by a 25-watt frosted bulb placed centrally over the maze, near the top of the chamber. Before each daily trial, the animals were placed in a cage directly under the center of the maze table for one hour, and were carried in the same direction in being transferred from this cage to the entrance box. Two rats from Group I and two from Group II were tested. The incentive was whole-wheat bread. Observations were

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made through a small peep hole, 0.75 inches in diameter. The north and south position of the food box was varied irregularly in accordance with the plan indicated in the first column of Table 3. The results covering the first 20 trials, and two later trials in which the maze was rotated 90° are also given in Table 3. The animals formed marked position habits as might have been predicted. It appeared altogether unlikely that these would be broken up without resorting to punishment. Since the application of punishment did not seem feasable at the time, the problem was discontinued.

REFERENCES

1. CARR, H. Maze studies with the white rat: I. Normal animals. *J. Anim. Behav.* 1917, **7**, 259-275.
2. DASHIELL, J. F. Some transfer factors in maze learning by the white rat. *Psychobiol.*, 1920, **2**, 329-350.
3. ————. The need for analytical study of the maze problem. *Psychobiol.*, 1920, **2**, 181-186.
4. HUBBERT, H. B. & LASHLEY, K. S. Retroactive association and the elimination of errors in the maze. *J. Anim. Behav.*, 1917, **7**, 130-138.
5. PETERSON, J. Completeness of response as an explanation principle in learning. *Psychol. Rev.*, 1916, **23**, 153-163.
6. ————. The effect of length of blind alleys on maze learning: an experiment on twenty-four white rats. *Behav. Monog.*, 1917, **3**, No. 15. Pp. 53.
7. WARDEN, C. J. Some factors determining the order of elimination of culs-de-sac in the maze. *J. Exper. Psychol.*, 1923, **6**, 192-210.
8. ————. A standard animal maze for general laboratory use. *J. Genet. Psychol.*, 1929, **36**, 174-176.

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LE FACTEUR D'ORIENTATION GÉNÉRALE DANS L'APPRENTISSAGE DU LABYRINTHE PAR LE RAT BLANC

(Résumé)

Dans la première expérience on a étudié l'influence de la rotation et d'autres facteurs, introduits aprés l'apprentissage, sur le rendement dans le labyrinthe. On a trouvé que la rotation du labyrinthe seul, ou celle du labyrinthe et de la cage sur leur axe commune ont changé le lieu des erreurs dans le rendement suivant de sorte de suggérer que l'orientation générale précédente a été transférée à la situation nouvelle. On n'a fait aucune analyse relative à la nature de la tendance à l'orientation. La deuxième expérience a étudié l'influence du renversement d'un type spécial de labyrinthe Y sur le lieu des erreurs. Dans cette situation simple l'effet du premier labyrinthe n'a pas été transféré d'une manière définie au deuxième.

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La troisième expérience a compris un essai préliminaire d'entraîner le rat blanc à tourner constamment au nord mais il a fallu abandonner ceci à cause de la fixité des habitudes de position.

WARDEN ET FOX

DER FAKTOR DER ALLGEMEINEN ORIENTIERUNG IN DER BEHERRSCHUNG VON LABYRINTHEN DURCH WEISSE RATTEN

(Referat)

Im Hauptversuch wurde der Einfluss der Rotation und anderer, nach dem Lernen eingeführter Faktoren auf das Benehmen im Labyrinth untersucht. Es wurde festgestellt, dass Rotation des Labyrinthes allein, oder des Labyrinthes und des Käfigs um ihre gemeine Achse, den geometrischen Pfad ('locus') der späteren Fehler auf eine Weise verschob, welche andeutete, dass die frühere allgemeine Orientierung auf die neue Lage übertragen wurde. Eine Analyse bezüglich der Natur der orienterenden Tendenz wurde nicht gemacht. Der zweite Versuch befasste sich mit dem auf den geometrischen Pfad der Fehler ausgeübten Einfluss, wenn eine besondere Sorte von Y-Labyrinth umgekehrt wurde. In dieser einfachen Situation wurde der Effect aus dem ersten Labyrinth nicht auf sehr bestimmte Weise in das zweite übertragen. Der dritte Versuch umfasste einen vorbereitenden Versuch, weisse Ratten so zu dressieren, dass sie sich beständig dem Norden zuwendeten. Dieser Versuch musste aber wegen der Festigkeit von Stellungsgewohnheiten ('position habits') aufgegeben werden.

WARDEN UND FOX

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